Introduction

A heatmap displays a matrix as a grid of coloured cells, with numerical values encoded by colour. The geom is a workhorse across genomics (gene-expression matrices), clinical research (cross-tabulations of conditions and measurements), classifier evaluation (confusion matrices), and exploratory data analysis (correlation matrices). Two design choices dominate readability: the colour palette (sequential, diverging, or qualitative) and the row / column order, which often comes from hierarchical clustering on rows, columns, or both.

Prerequisites

A working understanding of matrices, hierarchical clustering, and colour palette selection (sequential vs diverging).

Theory

Two ecosystems handle heatmaps in R:

- **`ggplot2` + `geom_tile()`**: flexible but requires long-format data and manual handling of clustering. Best when the heatmap is one panel in a larger composition.
- **`pheatmap` and `ComplexHeatmap`**: dedicated packages that handle clustering, dendrograms, row/column annotations, and z-scoring natively. Best when the heatmap is the figure and customisation focus is on annotation tracks.

The colour scale must match the data semantics: sequential (`viridis`, `Blues`) for ordered values without a meaningful zero, diverging (`RdBu`, `BrBG`) for z-scored data or any quantity centred on zero. Heatmaps with a one-sided palette applied to centred data are misleading because the visual midpoint does not match the numerical midpoint.

Assumptions

A matrix or matrix-like structure with rows and columns that admit a sensible reordering. For clustering, rows and columns must be on comparable scales (or pre-scaled per row).

R Implementation

```r
library(ggplot2)

# ggplot tile heatmap
d <- as.data.frame(scale(as.matrix(mtcars)))
d$car <- rownames(d)
d_long <- reshape2::melt(d, id.vars = "car")
ggplot(d_long, aes(variable, car, fill = value)) +
  geom_tile() +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red",
                       midpoint = 0) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# pheatmap with hierarchical clustering
library(pheatmap)
pheatmap(scale(mtcars), clustering_method = "ward.D2",
         color = colorRampPalette(c("blue", "white", "red"))(100))
```

Output & Results

Two heatmaps of the z-scored `mtcars` matrix: a hand-built `ggplot2` tile heatmap (no clustering, alphabetical row order) and a `pheatmap` version with Ward's hierarchical clustering on both rows and columns plus accompanying dendrograms. The clustered version reveals two clear groups of cars distinguished by their efficiency-versus-power profile; the unclustered version requires the reader to find that structure unaided.

Interpretation

A reporting sentence (figure caption): "Heatmap of z-scored vehicle attributes (n = 32 cars × 11 variables); colour encodes z-score on a blue-white-red diverging scale centred at zero, and rows and columns are clustered with Ward's method on Euclidean distance." Always state the scaling, the distance metric, and the linkage method; results depend on all three.

Practical Tips

- Centre and scale rows (z-score per row) before clustering whenever variables are on different scales; otherwise, large-magnitude variables dominate the distance metric.
- Use a diverging palette with an explicit midpoint for z-scored data; one-sided sequential palettes collapse the visual midpoint and obscure structure.
- For large matrices (thousands of rows or columns), `ComplexHeatmap` scales better than `pheatmap` and supports complex annotation tracks (categorical, continuous, mark-driven).
- Add side annotations (group, condition, batch) to help readers spot whether clustering tracks biology or technical artefacts.
- Cluster only the dimensions that should be ordered; force the other dimension's order when biological or temporal meaning matters.
- Use `cluster_rows = FALSE` (or `cluster_cols = FALSE`) when an existing ordering carries information you want preserved (e.g. chromosome position).

R Packages Used

`ggplot2` for `geom_tile()` and flexible composition; `pheatmap` for one-call heatmaps with clustering; `ComplexHeatmap` for large-scale heatmaps with rich annotations; `superheat` for an alternative ggplot-flavoured heatmap with linked summary plots; `viridis` and `RColorBrewer` for palette selection.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts